

PUFA stabilizes a conductive state of the selectivity filter in IKs channels

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Abstract

In cardiomyocytes, the KCNQ1/KCNE1 channel complex mediates the slow delayed-rectifier current (IKs), pivotal during the repolarization phase of the ventricular action potential. Mutations in IKs cause Long QT Syndrome (LQTS), a syndrome with a prolonged QT interval on the ECG, which increases the risk of ventricular arrhythmia and sudden cardiac death. One potential therapeutical intervention for LQTS is based on targeting IKs channels to restore channel function and/or the physiological QT interval. Polyunsaturated fatty acids (PUFAs) are potent activators of KCNQ1 channels and activate IKs channels by binding to two different sites, one in the voltage sensor domain (VSD) – which shifts the voltage dependence to more negative voltages– and the other in the pore domain (PD) – which increases the maximal conductance of the channels (Gmax). However, the mechanism by which PUFAs increase the Gmax of the IKs channels is still poorly understood. In addition, it is unclear why IKs channels have a very small single channel conductance and a low open probability or whether PUFAs affect any of these properties of IKs channels. Our results suggest that the selectivity filter in KCNQ1 is normally unstable, contributing to the low open probability, and that the PUFA-induced increase in Gmax is caused by a stabilization of the selectivity filter in an open-conductive state.

eLife assessment

This manuscript reveals an **important** mechanism of KCNQ1/IKs channel gating and PUFA modulation of this mechanism. This mechanism is supported by **convincing** single channel recordings, macroscopic current recordings and mutational analyses. These findings are of importance to the ion channel field and possibly future therapeutic applications.

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Introduction

The voltage-gated K⁺ channel KCNQ1, also referred to as Kv7.1, is expressed in the heart. Here the channel associates with the accessory KCNE1 subunit, generating the *so-called* slow delayed-rectifier (IKs) current, an important contributor to the repolarizing phase of the ventricular action potential (AP) ^{1,2,3}. Loss-of-function mutations of the KCNQ1/KCNE1 complex are associated with Long QT Syndrome (LQTS) ^{4,5}. These LQTS mutations cause a reduction of the channel current, leading to a significant prolongation of the ventricular AP waveform that can be seen on the electrocardiogram as a prolonged QT interval ³ (**Fig. 1A**). These channel mutations, or dysfunction, increase the risk of developing cardiac arrhythmias, which can lead to sudden cardiac death ⁶. At present, treatment of LQTS is mainly based on the usage of β -blockers (most used are long-lasting preparations such as nadolol and atenolol) and implantable cardioverter-defibrillators, especially for patients with high risk of sudden death and frequent syncope ⁷. However, both approaches do not shorten the QT interval duration. Restoring the physiological duration of the QT interval could be achieved by increasing the activity of the KCNQ1/KCNE1 channel complex ⁸ (**Fig. 1A**).

Like other Kv channels, KCNQ1 has a typical tetrameric structure of four α -subunits. Each α -subunit is composed of six transmembrane segments, with S1-S4 forming the voltage sensor domain (VSD) and S5-S6 forming the pore domain (PD) (**Figure 1D**). However, KCNQ1/KCNE1 channel complex has a very small single channel conductance and a very low open probability compared to other Kv channels ⁹. The mechanisms behind the small conductance and low open probability are not understood.

Polyunsaturated fatty acids (PUFA), in particular omega 3, are known to exert a protective effect on sudden cardiac death and are recommended in the diet at least twice a week ¹⁰. PUFAs have been shown to increase IKs currents by a dual mechanism of action, characteristically described as a “Lipoelectric mechanism” ¹¹. According to this mechanism, PUFAs can increase IKs currents by shifting the voltage-dependence of activation ($\Delta V_{0.5}$) toward negative voltages and increase the maximum conductance of the channel (ΔG_{max}) ^{12,13,14,15,16} (**Figure 1B-C**).

The two PUFA effects ($\Delta V_{0.5}$ and ΔG_{max}) on KCNQ1 are independent of each other and originate from the binding of PUFA to two different sites, conventionally indicated as Site I and Site II ^{14,18}. Site I is found in the VSD, where the PUFA head group interacts with the positive charges in the S4 segment and causes the channel to open at more negative potentials ¹⁹. Site II is found in the pore domain, where electrostatic interactions between the PUFA head group and the positively charged residue, K326, facilitate an increase in the maximal conductance of the channel ¹⁸ (**Figure 1E-F**).

Although previous studies have shown the importance of K326 for PUFA in increasing G_{max} through binding at site II ¹⁸, the mechanism by which PUFAs increase G_{max} is still not understood. In this study we use a combination of two-electrode voltage clamp, single channel recordings, site-directed mutagenesis, and recent cryoEM structures to gain insight into the molecular mechanism underlying the G_{max} effect following PUFA binding at site II. We propose a novel mechanism where the selectivity filter of KCNQ1 is normally unstable and the binding of PUFA to site II stabilizes a network of interactions at the selectivity filter, which in turn, leads to a more open and stable conformation of the KCNQ1 pore and thus an increase in channel open probability.

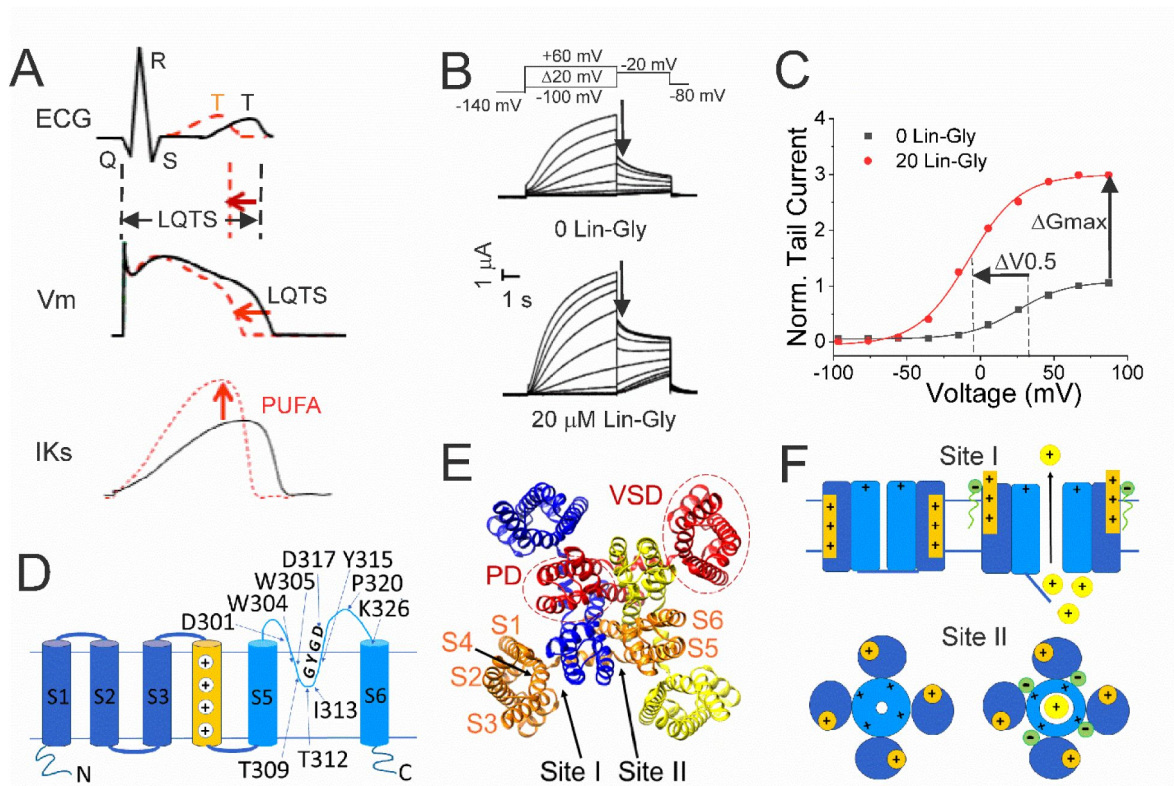


Figure 1.

PUFAs activate KCNQ1/KCNE1 channels.

A) (black) Prolonged QT interval in the ECG is due to for example loss-of-function mutations of KCNQ1/KCNE1 channels that generate the IKs current that normally contributes to the repolarizing phase of the ventricular AP. (red) PUFAs are potent activators of KCNQ1/KCNE1 channels that can restore the normal functioning of the channel and restore the AP duration and the QT interval. B) Representative current traces of KCNQ1/KCNE1 in 0 μ M and 20 μ M of Lin-Glycine. Voltage protocol on top. C) Conductance versus voltage curves from tail currents (measured at arrows in B). Channel activation by PUFA results in two main effects: a shift of the voltage-dependence of activation ($\Delta V_{0.5}$) and an increase in the channel maximum conductance (ΔG_{max}). D) KCNQ1 transmembrane topology. Residues mutated in this study are labeled. E) KCNQ1 top view (PDB: 6UZZ) with PUFA binding sites: Site I, at the VSD; and Site II, at the pore domain. The four subunits are shown in four different colors. F) Cartoon of PUFA mechanism of action. Site I, top panel. Electrostatic interactions between PUFA head groups and positively charged residues in S4 facilitate channel activation by stabilizing the outward state of S4. Site II, bottom panel. PUFA interaction with residues in the pore domain facilitates the increase in the maximum channel conductance.

Material and methods

Molecular biology

KCNQ1 (UniProt: P51787), KCNE1 (Uniprot: P15382) cRNA were transcribed using the mMessage mMachine T7 kit (Ambion). KCNQ1 was co-expressed with KCNE1 subunit, following a 3:1, weight: weight (Q1:E1) cRNA ratio to make up the KCNQ1/KCNE1 currents. Site-directed mutagenesis was performed using the Quickchange II XL Mutagenesis Kit (QIAGEN Sciences) for mutations in KCNQ1. 50 ng of complementary RNA was injected into defolliculated *Xenopus laevis* oocytes (Ecocyte Bioscience, Austin; Xenopus 1 Corp, Dexter) for channel expression. After RNA injection, cells were incubated for 24–96 h, in standard ND96 solution (96 mM NaCl, 2 mM KCl, 1 mM MgCl₂, 1.8 mM CaCl₂, 5 mM HEPES, 1M sodium pyruvate and penicillin (10,000 units)-streptomycin (10 mg/ml in 0.9% NaCl; pH 7.5) at 16°C before experiments.

Two-electrode voltage clamp

KCNQ1/KCNE1 currents were recorded using the two-electrode voltage clamp (TEVC) technique. The recording chamber was filled with ND96 solution (96 mM NaCl, 2 mM KCl, 1 mM MgCl₂, 1.8 mM CaCl₂, and 5 mM HEPES; pH 7.5). Pipettes were filled with 3M KCl. The polyunsaturated fatty acid (PUFA) Lin-Glycine was purchased from Cayman Chemicals (Ann Arbor, MI), kept in stock of 100 mM with ethanol at -20°C and diluted in ND96 solution the day of the experiments. Electrophysiological recordings were acquired using Clampex 10.7 software (Axon, pClamp, Molecular Devices). Lin-Glycine was perfused into the recording chamber in a pre-application step until reaching steady state, followed by an (I-V) protocol to measure the current-voltage relationship before and after perfusion. During PUFA application, cells were held at -80 mV and depolarization from -80 mV to 0 mV (5-s) was applied every 30 s, before stepping to -40 mV and then back to -80 mV. In the I-V protocol, a hyperpolarized step from -80 mV to -140 mV is applied before 20-mV voltage steps from -100 mV to +60 mV. Tail currents are recorded at -20 mV before returning to holding potential.

Single channel recordings

Single channel currents were recorded from transiently transfected mouse *ltk*-fibroblast cells (LM cells) using 1.5 µL Lipofectamine 2000 (Thermo Fisher Scientific). Cells were transfected with 1.5 µg of pcDNA3 containing a linked KCNE1-KCNQ1 construct ²⁰[\[20\]](#), to ensure fully KCNE1-saturated complexes, in addition to a plasmid containing green fluorescent protein (GFP) to identify transfected cells. Cells were recorded from 24–48 hours after transfection using an Axopatch 200B amplifier, a Digidata 1440A and pClamp10 software (Molecular Devices, San Jose CA, USA). The bath solution contained (in mM): 135 KCl, 1 MgCl₂, 1 CaCl₂, 10 HEPES, 10 Dextrose (pH 7.4 with KOH). The pipette solution contained (in mM): 6 NaCl, 129 MES, 1 MgCl₂, 5 KCl, 1 CaCl₂, 10 HEPES (pH 7.4 with NaOH). Pipettes were pulled from thick-walled borosilicate glass (Sutter Instruments, Novato, CA, USA), using a linear multistage electrode puller (Sutter Instruments), fire polished and coated with Sylgard (Dow Corning, Midland, MI, USA). Electrode resistance was between 40 and 80 MΩ after polishing. Currents were sampled at 10 kHz, low-pass filtered at 2 kHz at acquisition and subsequently digitally filtered at 200 Hz for presentation and analysis. Data were collected using cell-attached patch configuration to minimize disruption to the patch and avoid rundown problems due to the loss of PIP₂. Lin-Glycine was solubilized in DMSO and added directly to the bath. Only patches that were largely free of endogenous currents and had few channels, such that there were several blank sweeps to average for use for leak subtraction, were analyzed.

Data analysis

To measure the conductance vs voltage (G-V) curve, KCNQ1/KCNE1 tail currents were measured at -20 mV and obtained values were plotted against the activation voltages and fitted to a Boltzmann function:

$$G(V) = G_{min} + (G_{max} - G_{min}) / \{1 + \exp[(V_{50} - V)/s]\}$$

where G_{min} is the minimal conductance, G_{max} is the maximal conductance, V_{50} the midpoint (which describes the voltage at which the conductance is half the maximal conductance established from the fit), and s is the slope of the curve in mV. The difference in G_{max} effect, before and after application of Lin-Glycine in each oocyte is used as a measure of the change in maximal conductance. To understand the concentration dependence of LIN-Glycine effect on G_{max} , the following concentration-response curve was fitted to the data:

$$G_{max}/G_{max}^0 = 1 + B / [1 + ([PUFA]_{50}/[PUFA])^H]$$

where B is the maximum relative increase in G_{max} $\{(G_{max} - G_{max}^0)/G_{max}^0\}$, $[PUFA]_{50}$ the PUFA concentration needed to cause 50% of the maximal effect, and H the Hill coefficient. Average values are expressed as mean $\pm$ SEM and n represents the number of experiments (unless specified).

Statistical analysis was conducted using GraphPad Prism 8 (GraphPad Software, Boston, Massachusetts, USA). Statistical tests used were One-way ANOVA with Dunnett's post-hoc multiple comparisons test (Fig. 5 [↗](#)) and student t test for all other single comparisons.

Data were analyzed using Clampfit 10.7 (pCLAMP), Origin Pro (OriginLab Corporation) and GraphPad Prism 8 software (GraphPad Software, Boston, Massachusetts, USA).

Results

Lin-Glycine drastically increases the chance of KCNQ1/KCNE1 channel opening

Lin-Glycine has been shown to increase the G_{max} in whole-oocyte recordings of KCNQ1/KCNE1 channels 2.5-fold ¹³[↗](#). To better understand how Lin-Glycine increases G_{max} , we here extended our analysis to the single-channel level to study the behavior of KCNQ1/KCNE1 in the absence and presence of Lin-Glycine (20 μ M) (Fig. 2 [↗](#)). Representative traces (Fig. 2A-B [↗](#)) and all points histograms (Fig. 2C-D [↗](#)) suggest that there is no change in single channel conductance between KCNQ1/KCNE1 in control and with Lin-Glycine. This shows that the G_{max} effect is not due to an increase in the single channel conductance. However, Lin-Glycine caused a decrease in the latency to first opening (In control, 1.78 ± 0.36 s, $n = 41$ sweeps and after 20 μ M Lin-Glycine, 1.03 ± 0.05 s, $n = 99$ sweeps; $p = 0.0025$) and an increase in the number of non-empty sweeps (Suppl. Fig. S1A [↗](#)). In control, there are a high number of empty sweeps as shown before for KCNQ1/KCNE1 channels ¹⁷[↗](#). The average current during the single channel sweeps was increased by 2.3 ± 0.33 times ($n = 4$ patches, $p = 0.006$) by the application of Lin-Glycine (Fig. 2E [↗](#)). The number of non-empty sweeps increased 2.85-fold (2.85 ± 0.85 , $n = 200$ sweeps from 4 patches; $p = 0.001$), going from a total of 41/200 non-empty sweeps in control to 99/200 when Lin-Glycine was applied. In contrast, once the channel opens, it seems to behave very similarly in control and Lin-Glycine because, in non-empty sweeps, the open probability (P_o) in the last second of the traces was almost identical between control and Lin-Glycine conditions ($P_o = 0.78 \pm 0.02$ ($n = 8$ sweeps) in control vs 0.87 ± 0.04 ($n = 8$ sweeps) in Lin-Glycine) (Fig. 2F [↗](#)). This high open probability was maintained in control

conditions even for longer depolarizing voltage pulses, as if once the channels had opened they stayed open for the remaining of the voltage steps (**Suppl. Fig. S2**). The 2.85-fold increase in the number of non-empty sweeps is very similar to the G_{max} increase seen in macroscopic currents.

The decrease in first latency is most likely due to an effect of Lin-Glycine on Site I in the VSD and related to the shift in voltage dependence caused by Lin-Glycine. In contrast, the increase in the number of non-empty sweeps is most likely an effect on Site II in the pore and related to the G_{max} effect. We conclude that the G_{max} effect of Lin-Glycine on KCNQ1/KCNE1 is mainly due to an increase in the P_o by increasing the number of non-empty sweeps.

Crevice residues affect PUFA ability to increase G_{max}

It was previously shown that PUFAs binding at Site II electrostatically interact with the positively charged residue K326, located just outside the selectivity filter¹⁸. We tested whether another residue very close to K326, the aspartic acid at position 301, is important for the PUFA interaction at Site II. Electrophysiological analysis revealed that when KCNQ1_WT/KCNE1 was mutated to KCNQ1_D301E/KCNE1, the ΔV_{0.5} was shifted similarly to the WT channel (**Supplementary Figure S3**) but a dramatic decrease of the G_{max} effect of Lin-Glycine was observed, showing the importance of this residue for PUFAs interaction at Site II (**Figure 3A-C**). However, it is not clear how PUFA binding at Site II increases G_{max}.

In our previous MD simulations¹⁴ based on the cryoEM structure of KCNQ1 with S4 in the activated state, PUFA binds to residues K326 and D301 at Site II that delimit a narrow crevice (**Figure 3D-E**), where PUFA could fit with both the head and the tail (**Figure 3D-E**). However, in the recent cryoEM structure of KCNQ1 with S4 in the resting state²¹ the crevice between K326 and D301 is now so narrow that PUFA seems unable to fit into it (**Figure 3F**). In addition, there are large rearrangements in the selectivity filter when comparing the cryoEM structures with S4 activated (PDB: 8SIK) and S4 resting (PDB: 8SIN) (**Figure 4**). We have made a video showing the changes and the reorganization of the pore between these two structures (Supplement Video S1). The conformation of the pore significantly varies between what seems like a conductive (PDB: 8SIK) and a non-conductive (PDB: 8SIN) state of the channel. For example, in the structure with S4 in the resting state, Y315 and D317 have swung outwards and left their positions in which they made hydrogen bonds with two tryptophans, W304 and W305, respectively. Y315 belongs to the K⁺ channels signature sequence, a stretch of 8 amino acids including TXXTXGYG, highly conserved among K⁺ channels of different families and D317 is just next to this sequence^{22, 23}. W304 and W305 belong to the aromatic ring cuff (**Figure 4C-D**) that has been suggested to be important for the stability of the open state of the selectivity filter in Kv channels^{24, 25}.

In addition, P320, which in the structure with an activated S4 sits in between W304 and W305 from two subunits (**Figure 4C**), has also swung outwards in the structure with S4 in the resting state and exposed a gap between W304 and W305 in the aromatic cuff (**Figure 4D**, see also supplementary Video S2). The homologous proline in other Kv channels sits between the two tryptophans and stabilizes the aromatic ring cuff. **Supplementary Figure S4** and **S5** show the extent of the movement for each residue in the selectivity filter and nearby regions between the two conformations.

Based on our single channel data with many empty sweeps, we propose the hypothesis that the KCNQ1 selectivity filter is inherently unstable and transitions between the two conformations - one conductive state that generates the non-empty sweeps and one non-conductive state that generates the empty sweeps in the single channel data. We further propose that PUFA binding to residues at Site II stabilizes and promotes the conductive state of the pore. If PUFA binding to K326 and D301 promotes a series of interactions that stabilize the pore, then residues at the upper part of the selectivity filter, such as Y315 and D317, might be involved in those interactions that promote an open-conductive conformation of the channel.

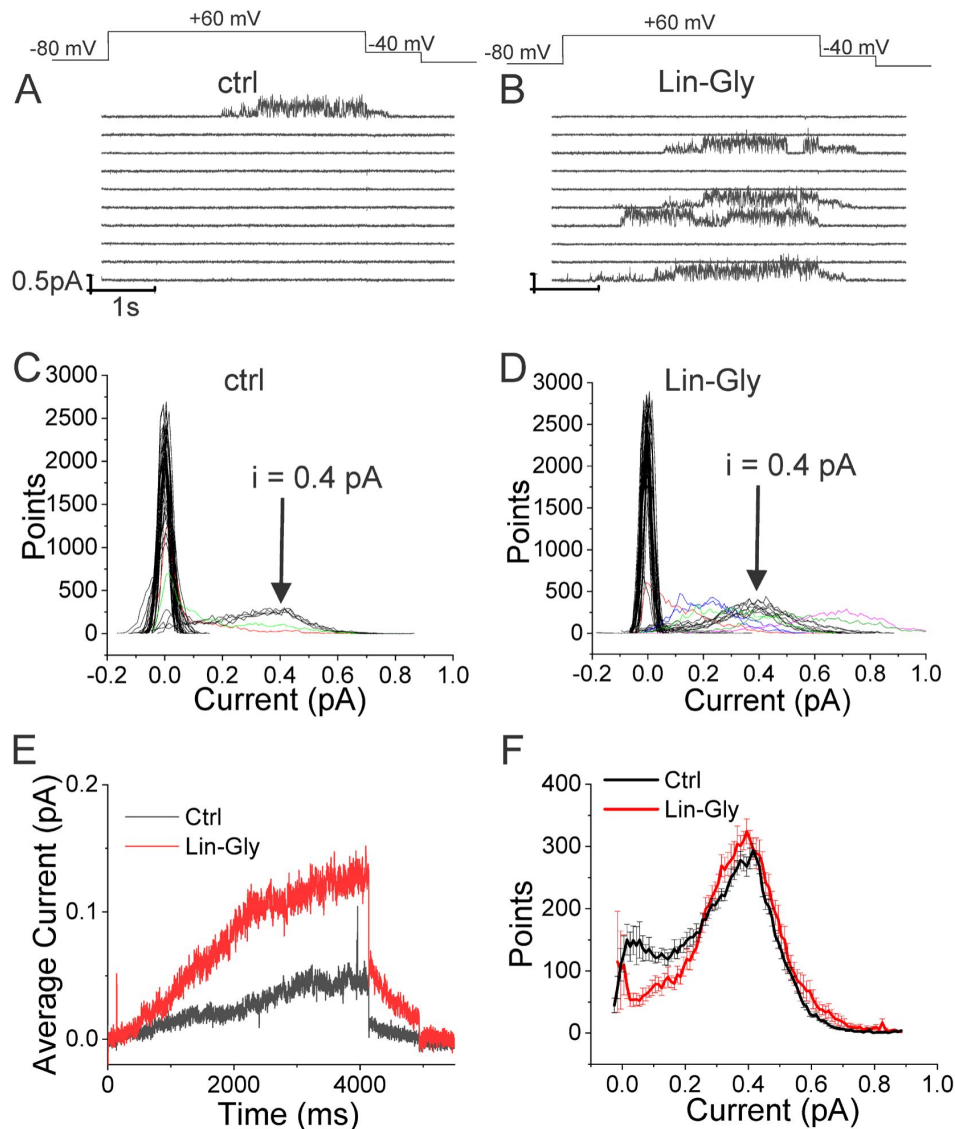
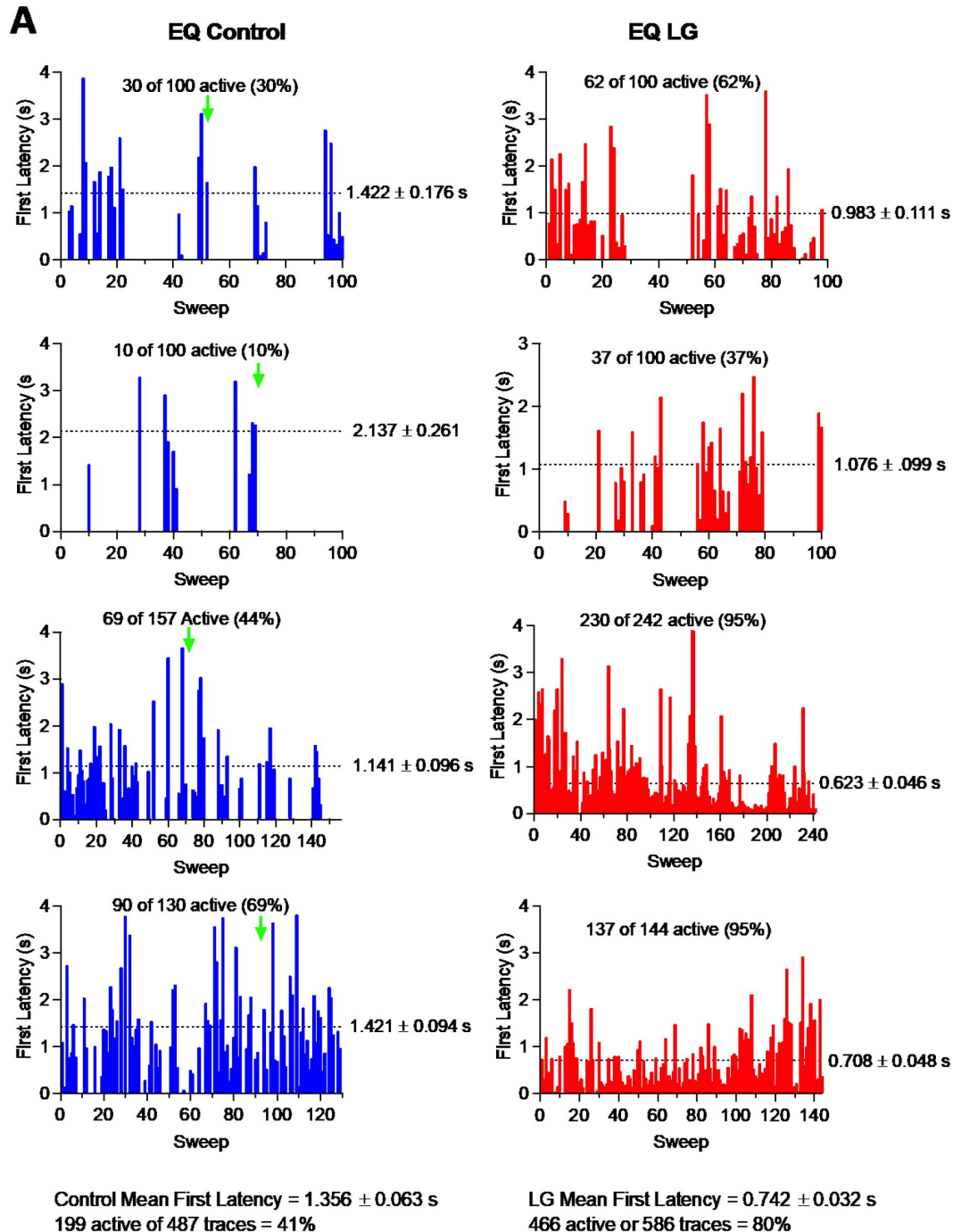


Figure 2.

Lin-Glycine induces an increase in the P_o of KCNQ1/KCNE1.

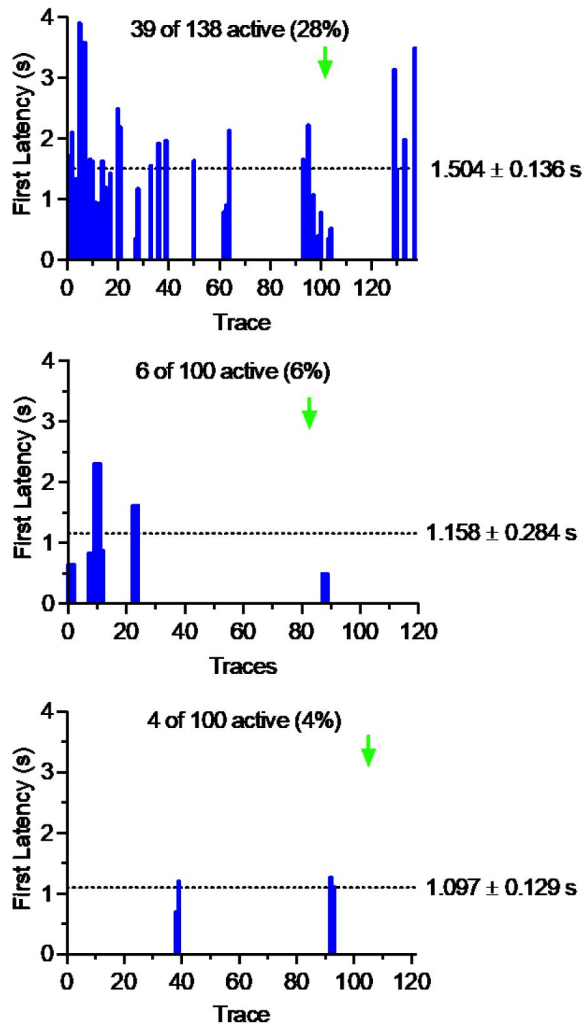
A-B) 10 consecutive traces of KCNQ1/KCNE1 in A) control and B) in the presence of Lin-Glycine (20 μ M) (top) Protocol used for the recordings. C-D. All-point amplitude histogram of 50 consecutive traces in C) control and D) Lin-Glycine. Note no change in the single-channel current amplitude, however an increase in the number of sweeps with channel opening is observed. Note that there were at least two channels in this patch. Different sweeps were assigned different colors to better visualize different types of channel behaviors. Note panels A-D are all from the same patch. E) Average currents of 100 sweeps in control and Lin-Glycine. F) All-point histogram of the last second of non-empty sweeps in control and in Lin-Glycine. We estimated the open probability from the all-point amplitude histogram by $P_o = \text{Sum}(iN/(i_{\text{estimate}}N_{\text{total}}))$, where N is the number of points for a specific current i in the histogram, $i_{\text{estimate}} = 0.4$ pA from the peak of the histogram, and $N_{\text{total}} = 10,000$ is the total number of points in the last second of the trace. $p = 0.78 \pm 0.02$ ($n = 8$ sweeps) and $p = 0.87 \pm 0.04$ ($n = 8$ sweeps) for Control and Lin-Glycine, respectively, from the same patch.



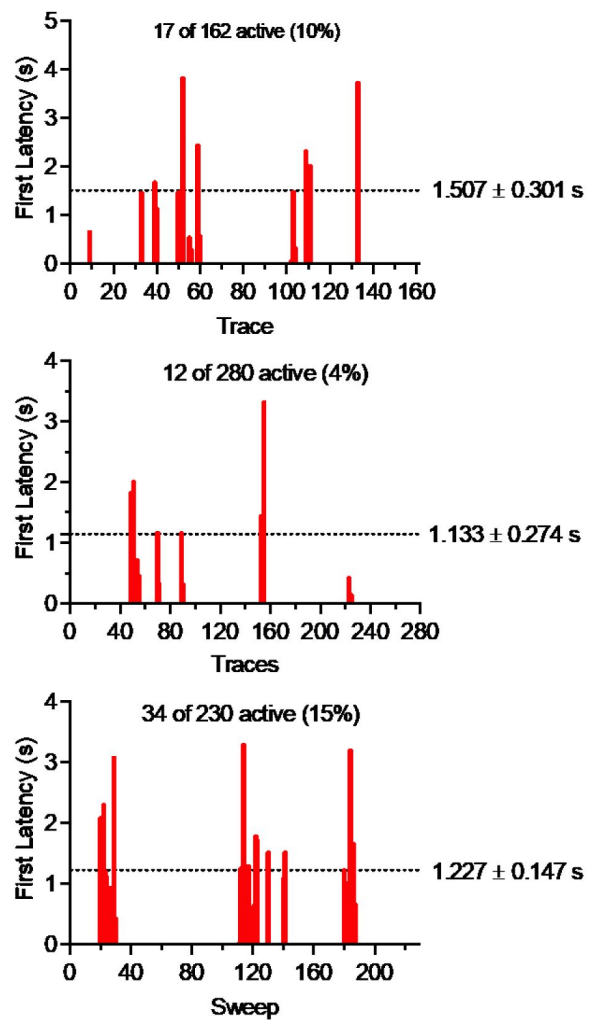
Supplementary Figure S1.

First latency to opening is shortened by Lin-Glycine.

A-B) Representative diary plot of first latency to opening during 4 s depolarizations to +60 mV for A) KCNQ1_WT /KCNE1 (EQ) and B) KCNQ1_Y315F/KCNE1 (Y315F) in control (left) and 20 μ M Lin-Glycine (right). Latency values of zero correspond to sweeps with no channel openings. Green arrow indicates point at which Lin-Glycine (LG) was added. Mean $\pm$ SEM first latency for each cell before and after Lin-Glycine (LG) are indicated next to each plot. Note that these are representative traces and for some patches there are more recordings than shown. In the analysis in the main text we used only traces recorded clearly before application of Lin-Gly and after Lin-Gly had been applied for several minutes, and we used only patches with low number of openings to avoid complications of simultaneous channel openings if more than one ion channel was present in the patch. Note that left and right panels in each row are from the same patch, but each row is from a different patch.

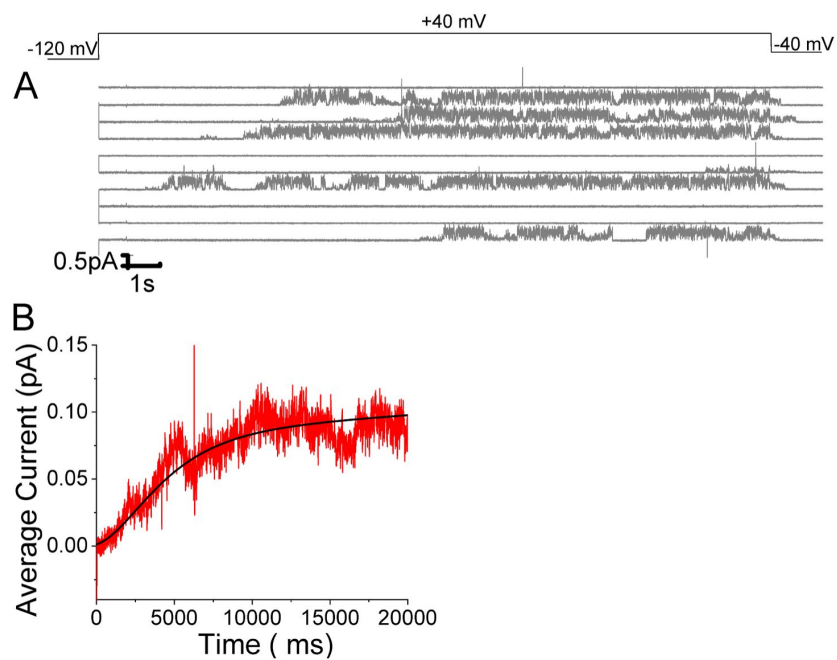
B**Control Y315F**

Control Y315F Mean First Latency = 1.480 ± 0.121 s
 49 active of 358 traces = 14%

LG Y315F

LG Y315F Mean First Latency = 1.336 ± 0.121 s
 60 active of 672 traces = 9%

Supplementary Figure S1. (continued)



Supplementary Figure S2.

Open probability stays high in KCNQ1/KCNE1 channels once opened.

A) 10 consecutive traces of KCNQ1/KCNE1 in response to 20-sec long voltage steps in control solutions. (top) Protocol used for the recordings. B) Average currents of 57 sweeps in control solution.

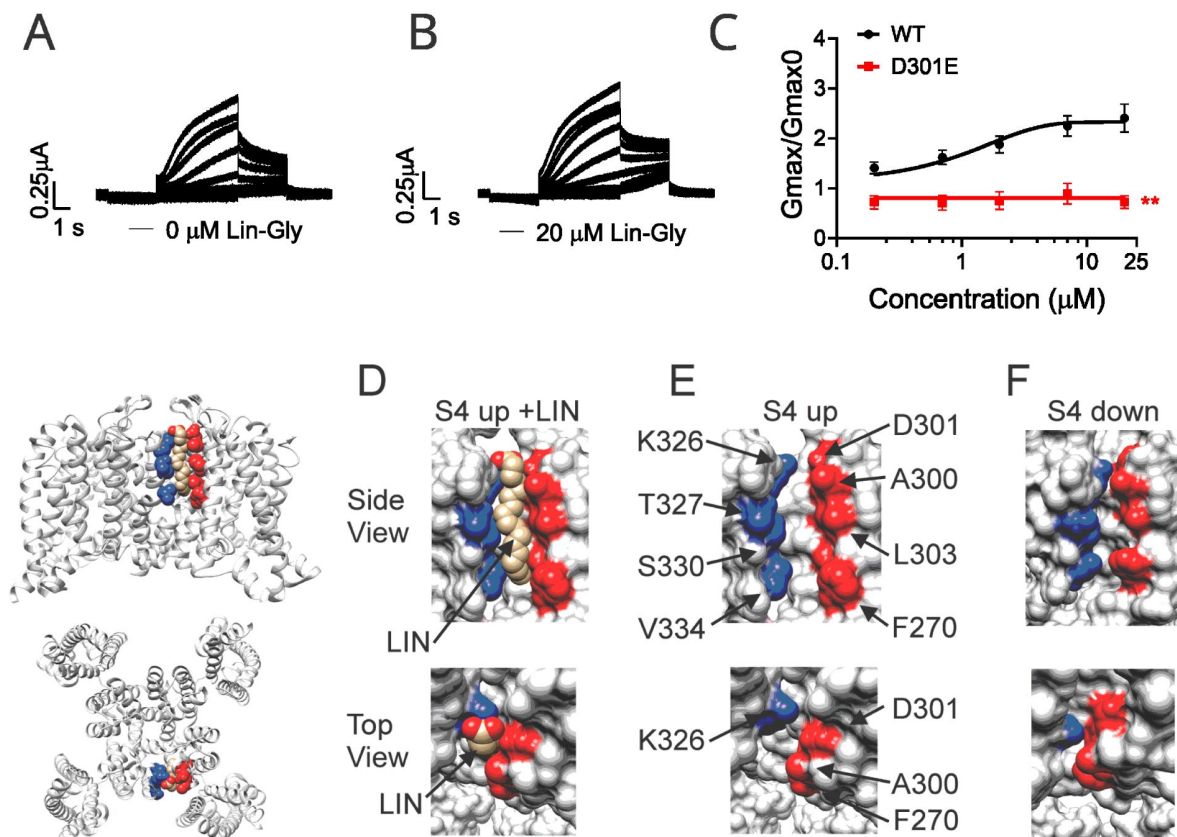


Figure 3.

PUFA binds to a state-dependent small crevice between K326 and D301.

A-B) KCNQ1_D301E/KCNE1 representative current traces A) in 0 μM of Lin-Glycine. B) After perfusion of 20 μM of Lin-Glycine. C) $G_{\text{max}}/G_{\text{max}0}$ for KCNQ1/KCNE1 channels and KCNQ1_D301E /KCNE1 channels. $G_{\text{max}}/G_{\text{max}0}$ was significantly reduced for the D301E mutation compared to WT channels ($p = 0.0018$, $n = 4$ oocytes). Student t test was used to conduct statistical analysis. D, E and F) Structures of crevice between S5 and S6 in KCNQ1 with S4 up (D and E) and S4 down (F). Residues that surround the crevice from S6 shown in blue (K326, T327, S330, V334) and from the pore helix and S5 in red (D301, A300, L303, F270). Remaining KCNQ1 residues shown in purple. On the left is shown the location of the crevice in KCNQ1 (top) side view and (bottom) top view. D) In MD simulations¹⁴, linoleic acid (LIN: gold color) fits in a narrow crevice present in the cryoEM structure of activated state KCNQ1 (S4 up). E) Same view as, in D but without LIN. F) In the cryoEM structure with S4 in the resting state (S4 down), the crevice between K326 and D301 is too narrow to fit LIN.

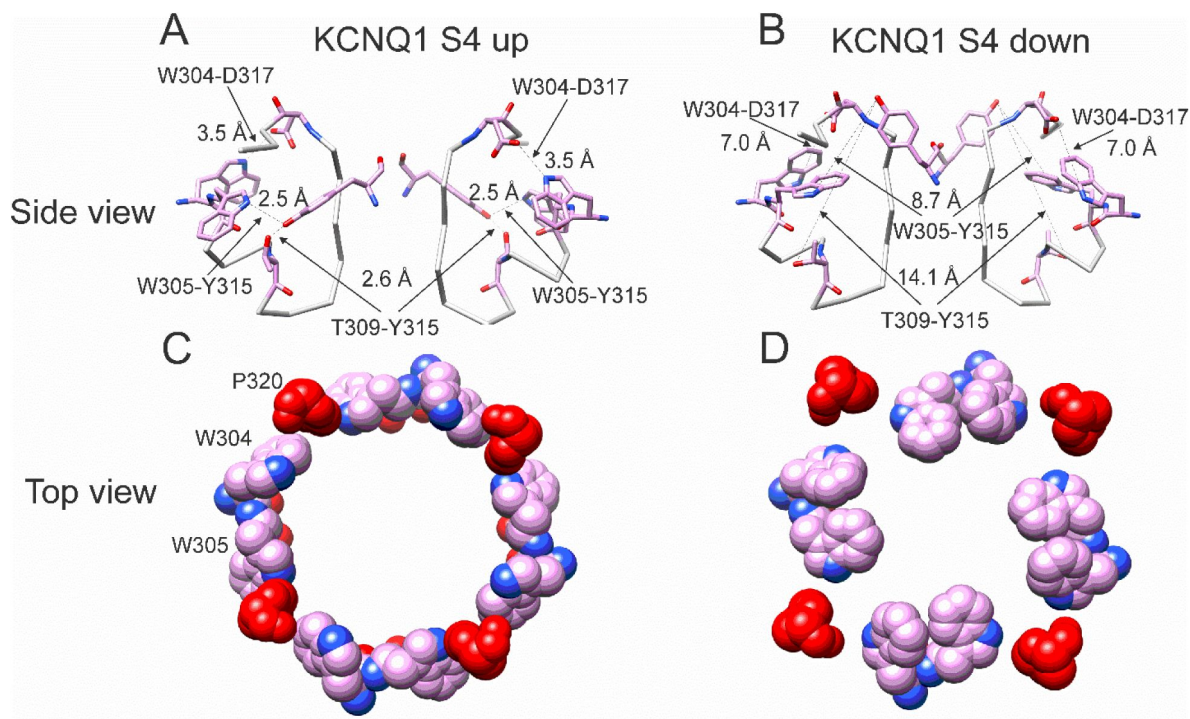
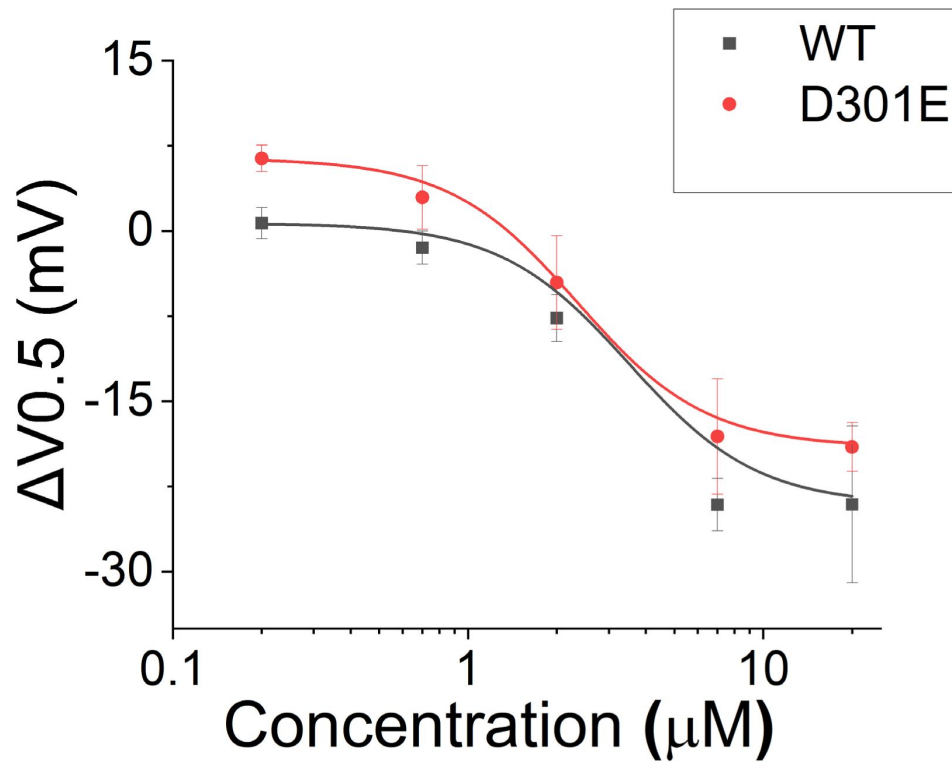


Figure 4.

Different conformations of selectivity filter in cryoEM structures with S4 activated or resting.

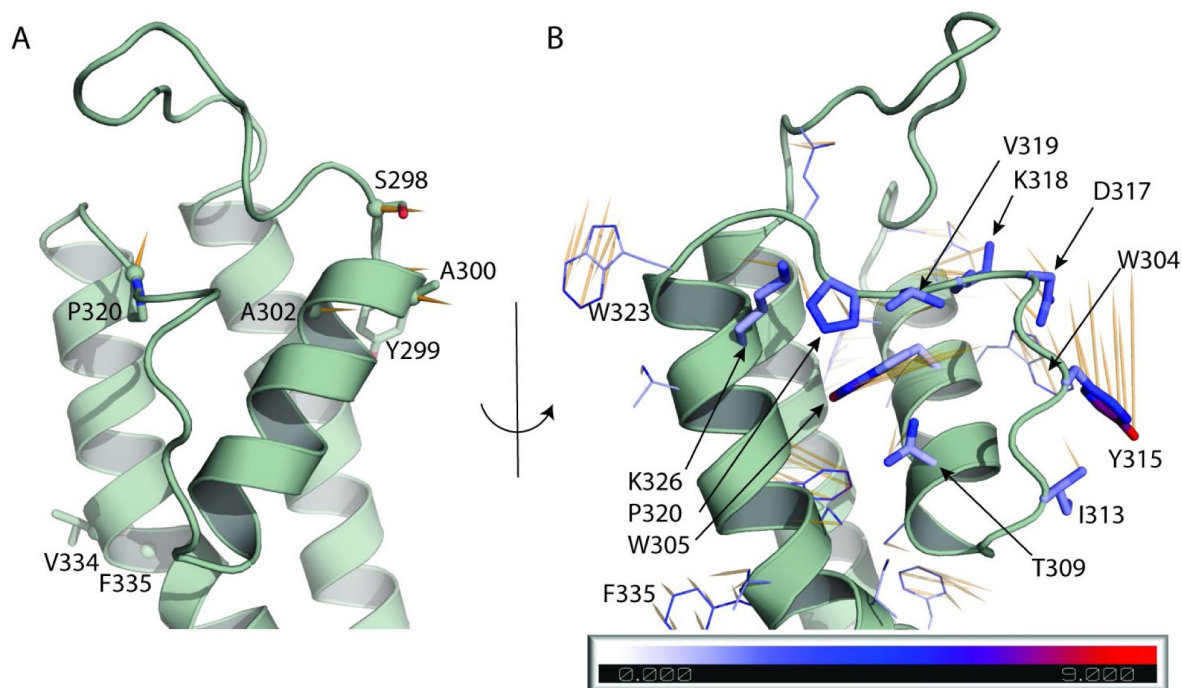
A) Selectivity filter of KCNQ1 with S4 activated (S4 up; PDB: 8SIK). Distances between D317 and W305 and between T309 and Y315 are short enough to form hydrogen bonds (dashed lines). B) Selectivity filter of KCNQ1 with S4 in resting state (S4 down; PDB: 8SIN). Distances between D317 and W305 and between T309 and Y315 are too long for hydrogen bonds (dashed lines). Only two subunits are shown for clarity. C-D). Aromatic cuff in KCNQ1 with C) S4 activated and D) resting S4. Note how P320 (red) moves away from its position in between W304 and W305 from two different subunits in the S4 down conformation.



Supplementary Figure S3.

Effect of Lin-Glycine in shifting the voltage-dependence of activation ($\Delta V_{0.5}$).

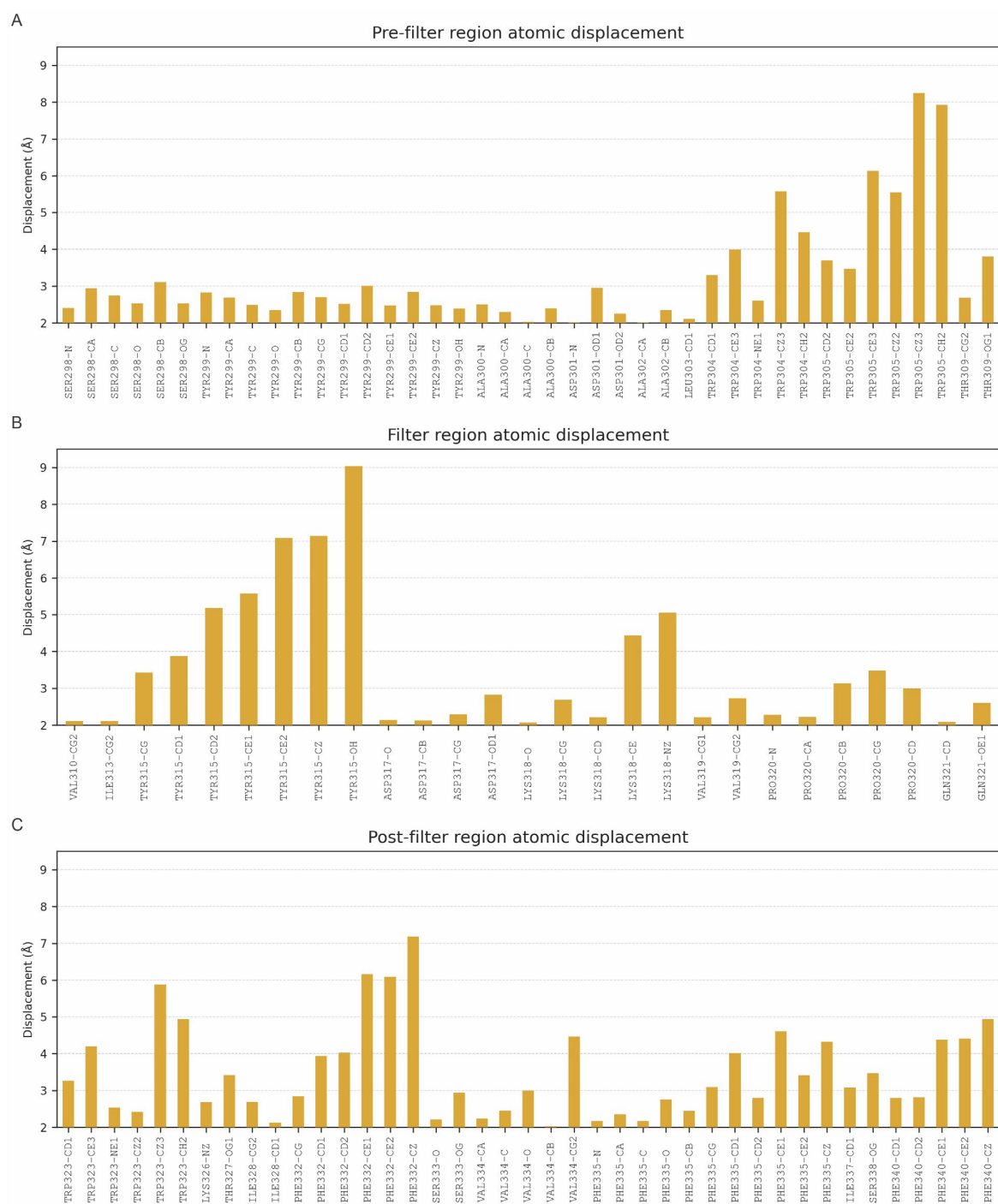
A similar shift in the voltage dependence of activation was found for KCNQ1_D301E/KCNE1 and KCNQ1/KCNE1 after perfusion of several concentrations of Lin-Glycine. Comparisons at 20 μ M of Lin-Glycine revealed no significant difference between the effect seen in KCNQ1/KCNE1 and KCNQ1_D301E ($p = 0.52$; $n = 3$ oocytes).



Supplementary Figure S4.

Atomic displacements at the selectivity filter and nearby regions in the transition from up to down state.

Each vector (shown in orange) starts from the position of a given atom in the up state (PDB ID 8SIK) and points to the position of the same atom in the down state (PDB ID 8SIN). **A.** Displacement of the Ca atoms of residues 298 to 341 is shown in orange if greater than 2 Å. The side chain of such residues is shown as sticks. **B.** Displacement of the side chains of residues 298 to 341. The side chain of residues W304, W305, T309, I313, Y315, D317, K318, V319, P320, K326 are highlighted as sticks and atoms are colored based on the atomic displacement values. Scale bar is in Å. In A and B, for clarity, only one monomer of the up state of KCNQ1 is shown.



Supplementary Figure S5.

Graphic indicating the atomic displacement of residues at the selectivity filter and nearby regions in the transition from up to down state.

Displacement of the Ca atoms of residues 298 to 341 is shown if greater than 2 Å. A) Displacement of atoms (Å) before the selectivity filter; B) residues of the selectivity filter; C) residues located just outside the selectivity filter region,

Residues that stabilize the selectivity filter are necessary for the G_{max} effect

If our pore stability hypothesis is correct, we should see altered effects of Lin-Glycine on G_{max} when residues important for the PUFA-promoted open-conductive conformation are mutated. We therefore mutated Y315F and D317E in KCNQ1. We tested Lin-Glycine on KCNQ1_Y315F/KCNE1 and KCNQ1_D317E/KCNE1 channels and compared the G_{max} effect to that of the WT channel. The effect of Lin-Glycine was reduced on both mutant channels, with a particularly dramatic decrease for Y315F (**Figure 5A**). For KCNQ1_Y315F/KCNE1 we compared the effect on the voltage-dependence of activation ($\Delta V_{0.5}$) and we observed a similar effect as for the WT channel (**Supplementary Figure S6**), with both showing an $\sim$ -25 mV shift. We also tested the Y315F mutation at the single channel level. As expected, Lin-Glycine did not increase the number of non-empty sweeps in KCNQ1_Y315F/KCNE1 channels (52/478 (10.9 % from 3 patches) of traces were non-empty in control and 44/533 (8.3% from 3 patches) of traces were non-empty in Lin-Gly) (**Fig. 6** and **Supplementary Figure 1B**). The mutation Y315F reduced the single channel current slightly from $I = 0.4$ pA in wildtype to $I = 0.3$ pA in the Y315F mutation (**Fig. 6 C-D**). The current average of all traces showed no increase in average current by Lin-Glycine (**Fig. 6 E**). This data shows that Y315 and D317E are necessary for the ability of Lin-Glycine to increase G_{max}. Conductance vs voltage curves (G-V) for WT and mutant channels are shown in **Supplementary Figure S7**.

We also found that another residue in the pore helix, T309, was important for the G_{max} effect of Lin-Glycine. The homologous residue in the Shaker channel was earlier suggested to be important for the stabilization of the selectivity filter by hydrogen bonding to one of the tryptophans in the aromatic ring²⁶. We generated the mutant channel KCNQ1_T309S/KCNE1 and measured the effect of Lin-Glycine. As shown in **Figure 5**, the effect of Lin-Glycine on G_{max} of the KCNQ1/KCNE1 mutant channel was noticeably reduced compared to the WT channel showing that this residue contributes to the G_{max} effect (**Figure 5A**). We also tested the involvement of proline, which makes up a part of the aromatic cuff in the activated state of the channel (**Fig. 4C**) by creating the mutant channel KCNQ1_P320L/KCNE1 and found a significant reduction of the G_{max} effect for this mutant (**Figure 5A**). All these data suggest that mutations of residues at the outer portion of the selectivity filter do affect the G_{max} increase by Lin-Glycine. These data are consistent with our hypothesis that PUFAs increase the G_{max} by affecting interactions that stabilize the selectivity filter.

The residues involved in the G_{max} effect are found near the external region of the selectivity filter, suggesting that the network of interactions that are altered during the switch from non-conductive to conductive state are confined to the outer region of the selectivity filter and pore helix of KCNQ1/KCNE1 channels. To test the specificity of the network localization, we mutated two residues, T312 and I313, in the internal portion of the selectivity filter. As a confirmation of our hypothesis, we found that for both KCNQ1/KCNE1 mutant channels, T312S and I313S, the effect of Lin-Glycine in increasing G_{max} resembled values obtained in the WT channels (2.49 ± 0.98 and 2.18 ± 0.05 , respectively) (**Figure 5B**). Mutations of residues in the more intracellular region of the selectivity filter do not affect the G_{max} increases, as if the interactions that stabilize the channel involve residues located near the external region part of the selectivity filter.

Taken together, our results suggest that the binding of PUFA to Site II increases G_{max} by promoting a series of interactions that stabilize the channel pore in the conductive state. For instance, we speculate that in the conductive state, hydrogen bonds between W304-D317 and W305-Y315, which are likely absent in the non-conductive conformation of KCNQ1, are created and that PUFA binding to K326 and D301 at Site II favors the transition towards the conductive state of the channel (**Figure 7**).

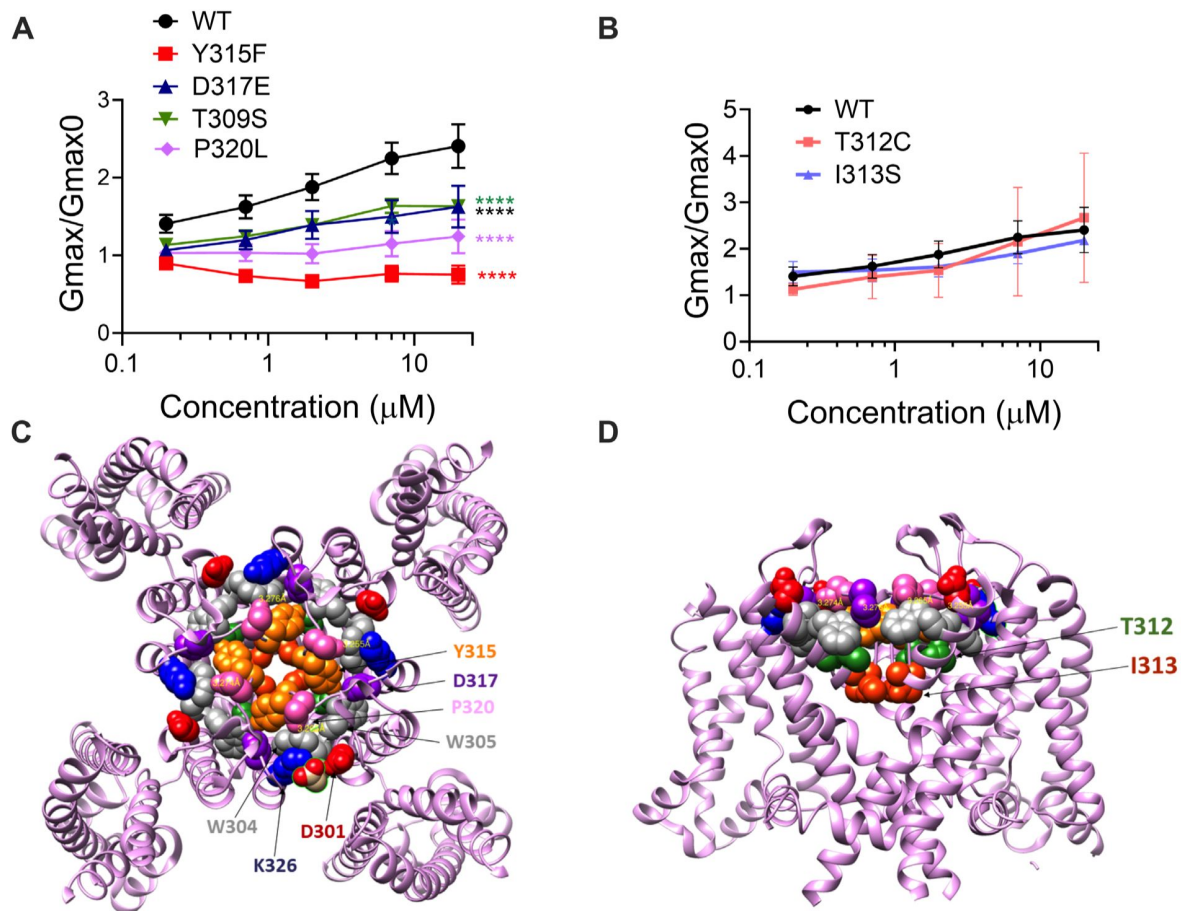


Figure 5.

The ability of Lin-Glycine to increase the channel conductance is reduced when channel pore residues are mutated.

A) G_{max}/G_{max0} values obtained for KCNQ1_WT/KCNE1 channel (black) and mutant channels. KCNQ1_Y315F/KCNE1 (red), KCNQ1_D317D/KCNE1 (blue), KCNQ1_T309S/KCNE1 (green) and KCNQ1_P320L/KCNE1 (purple), the G_{max}/G_{max0} is significantly reduced. ($p < 0.0001$ ($n = 4$ oocytes)). One-way ANOVA with Dunnett's post-hoc multiple comparisons test was used for statistical analysis. B) G_{max}/G_{max0} values for KCNQ1_T312C/KCNE1 and KCNQ1_I313S/KCNE1 ($p = 0.14$ and $p = 0.10$ respectively). ($n = 3$ oocytes). One-way ANOVA with Dunnett's post-hoc multiple comparisons test was used for statistical analysis. C-D) Top view and side view of KCNQ1 channel with mutated residues highlighted.

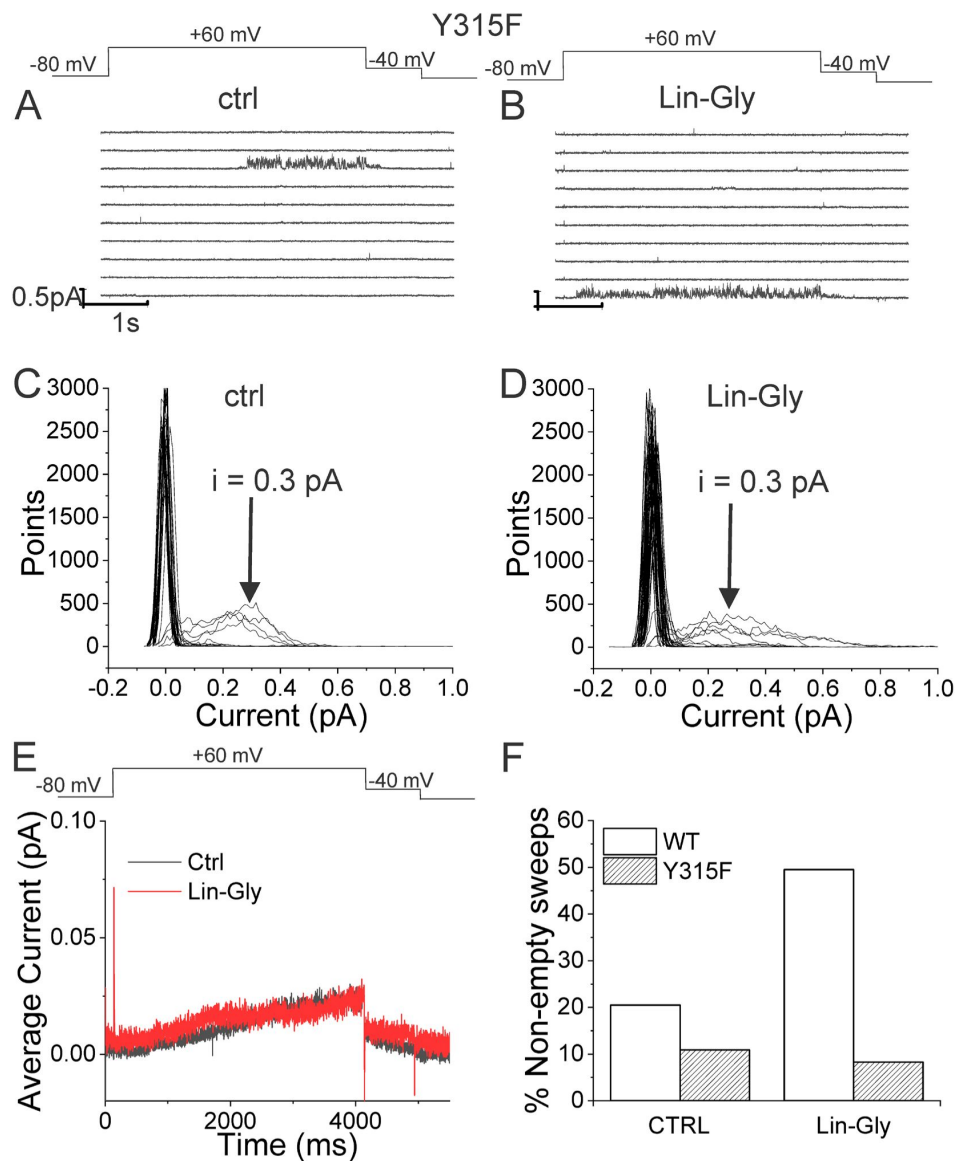
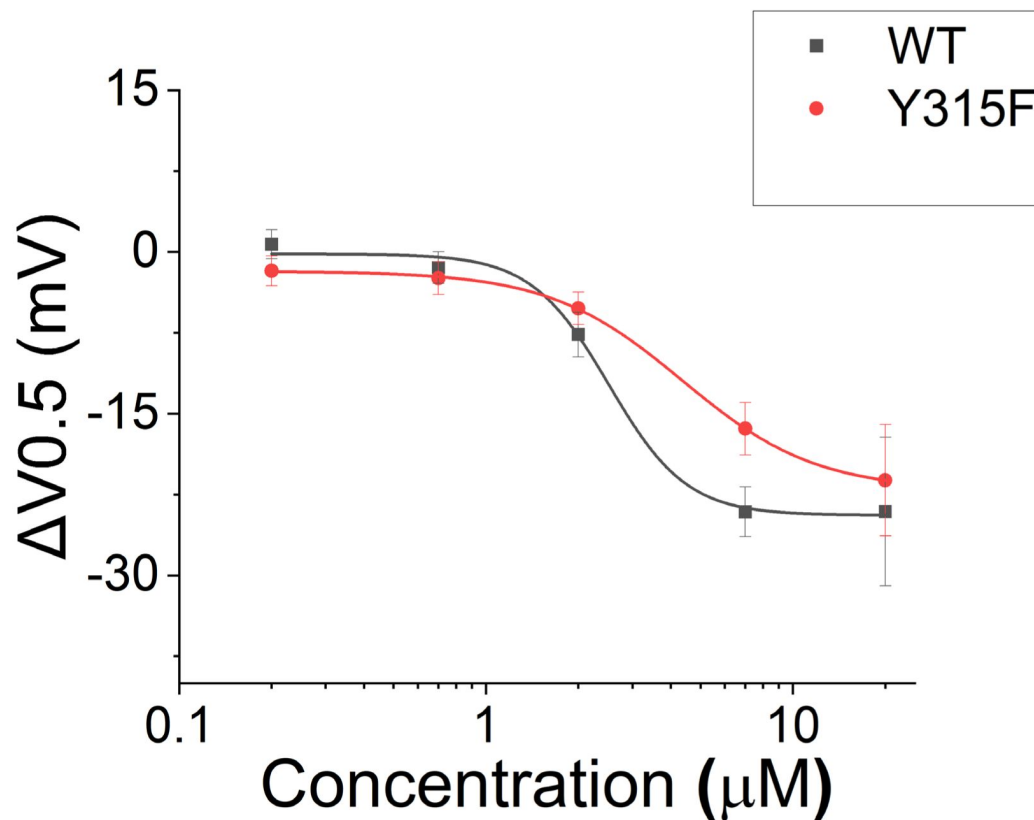


Figure 6.

Lin-Glycine does not increase the P_o of KCNQ1_Y315F/KCNE1.

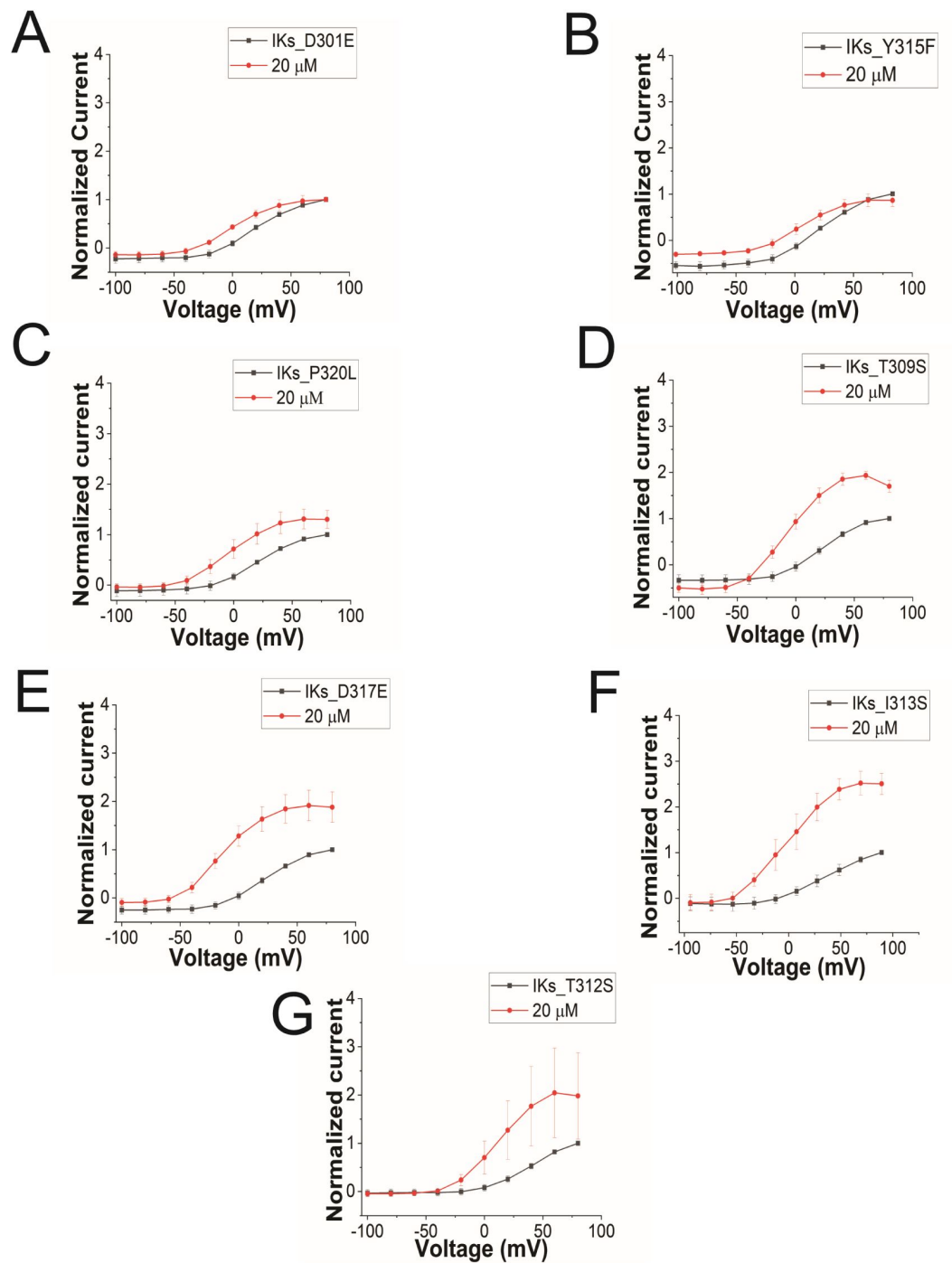
A-B) 10 consecutive traces of KCNQ1_Y315F/KCNE1 in A) control and B) in the presence of 20 μ M Lin-Glycine. (top) Protocol used for the recordings. C-D) All-point amplitude histogram of 50 consecutive traces in C) control and D) Lin-Glycine). The single-channel current amplitude was reduced to 0.3 pA compared to 0.4 pA for WT KCNQ1/KCNE1 (cf. **Figure 2C-D**). Note that there were at least two channels in this patch. E) Average currents of 478 sweeps in control and 533 sweeps in Lin-Glycine. Note that panels A-D were all from the same patch.



Supplementary Figure S6.

Effect of Lin-Glycine in shifting the voltage-dependence of activation ($\Delta V_{0.5}$).

A similar effect in shifting the voltage-dependence of activation was found for KCNQ1_Y315F/KCNE1 and KCNQ1/KCNE1 after perfusion of several concentrations of Lin-Glycine. Comparisons at 20 μ M of Lin-Glycine revealed no significant difference between the effect seen in KCNQ1/KCNE1 and KCNQ1_Y315F ($p = 0.7435$; $n = 4$ oocytes).



Supplementary Figure S7.

GV curves for KCNQ1 mutants.

GV curves in control (black) and after perfusion of 20 μ M of Lin-Gly (red) for A) KCNQ1_D301E/KCNE1, B) KCNQ1_Y315F/KCNE1, C) KCNQ1_P320L/KCNE1, D) KCNQ1_T309S/KCNE1, E) KCNQ1_D317E/KCNE1, F) KCNQ1_I313S/KCNE1, and G) KCNQ1_T312S/KCNE1.

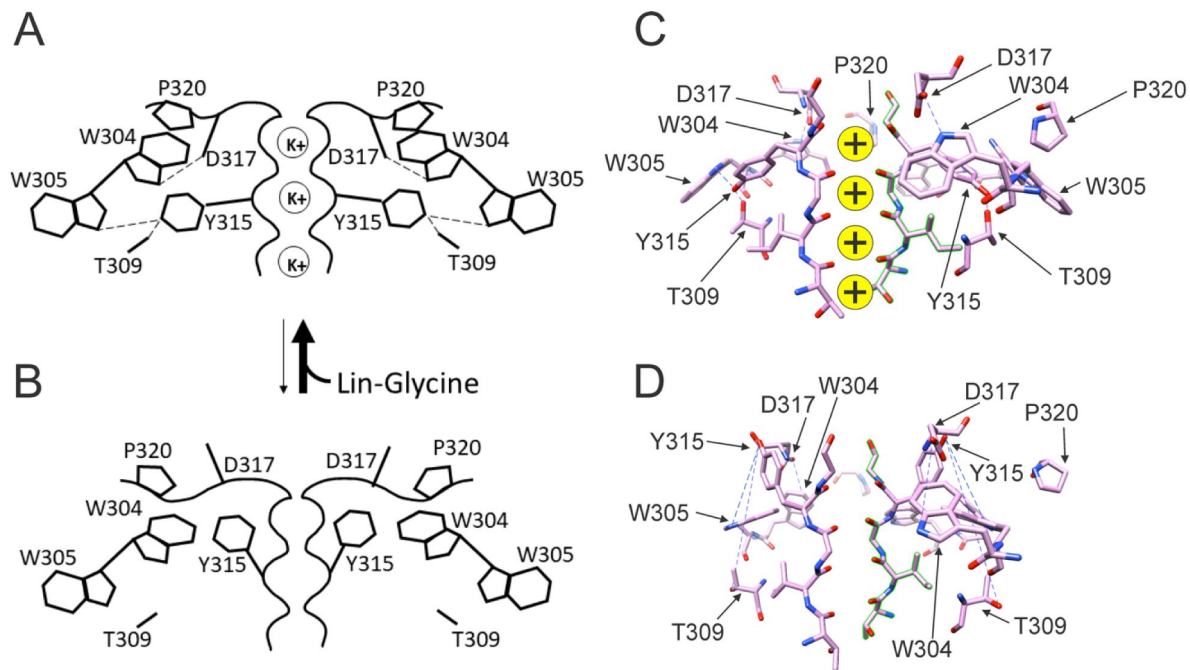


Figure 7.

Conformational changes occurring at the pore during the transitions between non-conductive and conductive states.

A) The binding of PUFA to site II between K326 and D301 induces a series of interactions between residues near the external part of the selectivity filter. W304-D317 and W305-Y315 form a hydrogen bond (dash line, black). Furthermore, Y315 interacts also with T309 (dash line, black). P320 is reoriented to sit on top of W304 and W305 to favor a more stable configuration of the aromatic ring cuff. The result of those new interactions is a more stable and conductive pore. B) In the non-conductive state those interactions are likely to be absent and this results in a more unstable selectivity filter. Also, P320 is now flipped from its position on top of the two tryptophan of the aromatic ring cuff. C) Cryo-EM selectivity filter with S4 in the activated-state representative of a conductive selectivity filter. D) Cryo-EM selectivity filter with S4 in the resting state, representative of a non-conductive selectivity filter.

Discussion

We have previously proposed models in which the effect of PUFAs on IKs channels involve the binding of PUFAs to two independent sites: one at the voltage sensor (Site I) and one at the pore domain (Site II). The result is a potent activation of IKs channels, with an increase of the maximum conductance by binding to Site II and a shift of the voltage dependence of activation of the channel by binding to Site I. The mechanism of the voltage shift effects at Site I, where the PUFA head group electrostatically interacts with positively charged S4 residues and thereby facilitates channel activation, has been investigated in previous studies^{14, 27, 18, 28}. However, less is known about the molecular mechanism by which PUFA increases the maximum conductance (G_{max}) of the channel following binding to Site II. A positively charged lysine at position 326 was suggested to be critical for PUFA-channel interaction, as mutations of K326 completely abolished the G_{max} effect¹⁸. We have here shown that the interaction at Site II also involves another residue, the aspartic acid 301, as mutagenesis analysis revealed that the effect of Lin-Glycine on G_{max} was abolished when the residue was substituted with a glutamic acid.

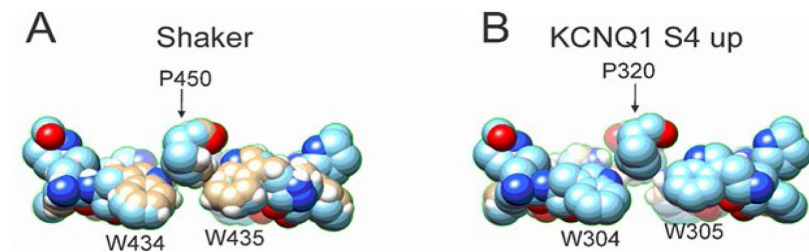
Our single channel recordings revealed that Lin-Glycine decreases the latency to first opening and increases the P_o. The decrease in the latency to first opening is most likely due to the binding of Lin-Glycine to Site I at the VSD and related to the shift in voltage dependence caused by Lin-Glycine (**Suppl. Fig S1**). In addition, our single channel recordings revealed that Lin-Glycine did not change the single channel conductance, but instead increased G_{max} by increasing the P_o of the channel. This increase in P_o was mainly due to an increase in non-empty sweeps when Lin-Glycine was applied in comparison to control solutions.

We tested the hypothesis that the effect of Lin-Glycine involved conformational changes in the selectivity filter following PUFA binding to two residues K326 and D301 at the pore domain. Those residues delimit a small crevice that seems to change in size in different structures with S4 up or down (**Figure 3, D-F**). Those residues seem to affect PUFA ability to increase G_{max}. We made several mutations of residues known to stabilize the selectivity filter in potassium channels (Y315F, D317E, T309S and P320L; **Fig. 5A**). The Lin-Glycine effect on the G_{max} was much reduced in these mutants suggesting that these residues are necessary for the G_{max} effect. To gain further insight into the molecular interactions that could underline the G_{max} effect by PUFA binding to site II, we used the latest KCNQ1 structure with the S4 segment in the resting state²¹. The selectivity filter in this structure showed a very different conformation of the pore region and selectivity filter of KCNQ1 compared to the structures with activated VSDs. Clearly, there will be other differences in the pore domain between structures with activated and resting VSDs, for example the state of the activation gate. Many of the interactions that have been shown to be important for the stability of the selectivity filter are missing in the KCNQ1 structure with a resting VSD. These changes in the selectivity filter can best be seen in our interpolation video between the states with the S4 segments moving from the resting state to the activated state (Supplement video 1). This gave us the idea that the effect of PUFAs in increasing the maximum conductance of the KCNQ1 channel is linked to their ability to stabilize the pore of the channel in a conductive state. It was previously shown that several interactions at the pore region of K⁺ channels are important for ensuring channel conductivity. For example, a feature conserved among K⁺ channels is the aromatic ring cuff that stabilizes the conducting state and plays a role in C-type inactivation^{24, 25, 26, 29}. This structure is made up of two tryptophan residues (W) and a proline residue (P) that sits in between the two tryptophan residues (Supplement video 2). In Shaker and KcsA K⁺ channels, the two tryptophans are involved in C-type inactivation and modification of those interactions manipulate the extent of inactivation^{30, 31}. In Shaker, breaking the hydrogen bonds between the two tryptophans of the aromatic ring cuff and the aspartic acid and tyrosine at the selectivity filter causes an acceleration of the rate C-type inactivation²⁶. In KCNQ1, the two tryptophans of the aromatic ring cuff correspond to W304 and W305. Another conserved residue

in K^+ channels important for the stability of the selectivity filter is P320 (equivalent to P450 in Shaker) that makes up part of the aromatic ring cuff. In the transition between the non-conductive and conductive state of KCNQ1, the proline pulls away from its position close to W304 and W305 and flips outwards (Supplement Video 2). We hypothesize that those hydrogen bonds between W304-D317 and W305-Y315 and the proline interactions are likely absent in the non-conductive conformation of KCNQ1, thereby generating the high number of empty sweeps in control conditions. Occasionally the channel will reform these bonds and interactions and transition into the conductive conformation, thereby generating the non-empty sweeps. We hypothesize that the binding of Lin-Glycine to K326 and D301 at site II biases the transition towards the conductive conformation, thereby increasing the number of non-empty sweeps and increasing G_{max} . We here show that mutations of residues involved in these hydrogen bonds in the selectivity filter greatly reduce or abolish the G_{max} effect of Lin-Glycine, as if Lin-Glycine increases the G_{max} by stabilizing a conductive state of the selectivity filter through these hydrogen bonds.

We noticed that the arrangement of the aromatic ring cuff is slightly different between Shaker K^+ channels and KCNQ1 channel. For instance, in Shaker, the proline residue sits in between the two tryptophan residues and stabilizes the aromatic ring cuff (**Supplementary Figure S8A** [↗](#)). In contrast, in KCNQ1 the proline is positioned a little further outward and away from the two tryptophan residues, thus generating a looser arrangement of the aromatic ring cuff (**Supplementary Figure S8B** [↗](#)). This difference might contribute to rendering the pore of KCNQ1 more unstable, resulting in a large number of empty sweeps and the characteristic flickering nature of channel openings [32](#) [↗](#). In Shaker channels the aromatic cuff seems more stable since it displays few empty sweeps and fewer flickering during bursts. This difference in the stability of the aromatic cuff between Shaker and KCNQ1 might explain why the effect of PUFA on G_{max} is large for KCNQ1 but not seen in Shaker [33](#) [↗](#), [34](#) [↗](#) even if the residues involved in the G_{max} effect are conserved between these two channels.

Our single channel data show that the KCNQ1/KCNE1 switches slowly (>10s seconds) between conductive and non-conductive states giving rise to many empty sweeps and a few active sweeps. However, once the channel is turned conductive during a depolarization, the channel stays conductive for the remainder of the sweep (See e.g. **Suppl Fig. S2** [↗](#) for 20-s sweeps), as if VSD activation stabilizes the conductive state. Therefore, we propose a model in which the selectivity filter is stabilized more in the non-conductive state when VSD is resting but stabilized more in the conductive state when VSD is activated. This model is consistent with the cryoEM structures of KCNQ1 with VSD resting displaying a non-conductive selectivity filter and with VSD activated displaying a conductive selectivity filter [21](#) [↗](#). The KCNQ1 structure with VSD resting was obtained in a low K^+ solution, which in other Kv channels would promote C-type inactivation of the selectivity filter. However, we think it is the resting state of the VSD, and not the low K^+ , that caused the non-conductive selectivity filter in the KCNQ1 structure, because a KCNQ1 structure with VSD activated in the low K^+ solution had a selectivity filter that was nearly identical to the KCNQ1 structure in high K^+ with VSD activated (except for fewer K^+ ions in the filter in low K^+ solutions) (See **Suppl. Fig S5 C** [↗](#) and **D** [↗](#) in [21](#) [↗](#)). Note also the currents from KCNQ1/KCNE1 channels do not have any external K^+ dependence (except for what is expected from changes in driving force) and that KCNQ1 channels are inhibited by high extracellular K^+ concentrations [35](#) [↗](#) in contrast to other Kv channels, such as Shaker K^+ channels, which are inactivated more in low K^+ [25](#) [↗](#), [29](#) [↗](#). Our studies suggest that the mechanism of PUFAs to increase KCNQ1/KCNE1 maximum conductance relies on the ability of PUFAs to favor the conductive conformation of the selectivity filter: PUFAs promote interactions between residues in the selectivity filter that stabilize the channel pore. Furthermore, we identify a crevice between K326 and D301 that is present in structures of KCNQ1 with activated VSD but not in structures where the VSD is in the resting state. We propose that PUFA binding in this crevice stabilizes the network of interactions that form the conductive form of the selectivity filter. The result is a more stable and conductive pore, which explains the increase in G_{max} by PUFAs.



Supplementary figure S8.

Comparison between the aromatic ring cuff configuration of Shaker and KCNQ1 channels.

A) In Shaker, P450 sits in between the two tryptophan and stabilizes the aromatic ring cuff. B) In contrast, in KCNQ1 the P320 is positioned further outward and away from the two tryptophan, generating a looser arrangement of the aromatic ring cuff.

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Reviewer #1 (Public review):

This study comes to an interesting conclusion: a polyunsaturated fatty acid, Lin-Glycine, increases the conductance of KCNQ1/KCNE1 channels by stabilizing a state of the selectivity filter that allows K⁺ conduction. The stabilization of a conducting state is well supported by single channel analysis, which shows that normally infrequent opening bursts occur more often in the presence of the PUFA. The linkage to PUFA action through the selectivity filter is supported by disruption of PUFA effects by mutation of residues which change conformation in two KCNQ1 structures from the literature. A definitive functional experiment is conducted by single channel recordings with selectivity filter domain mutation Y315F which ablates the Lin-Glycine effect on G_{max}. The computational exploration of two selectivity filter structures proposed to interact distinctly with Lin-Glycine is informative. Both mutation results and simulations converge on the proposed selectivity filter mechanism, although other possibilities for Lin-Glycine binding and action might be possible. Overall, the major claim of the abstract is well-supported: "... that the selectivity filter in KCNQ1 is normally unstable ... and that the PUFA-induced increase in G_{max} is caused by a stabilization of the selectivity filter in an open-conductive state."

<https://doi.org/10.7554/eLife.95852.3.sa3>

Reviewer #2 (Public review):

Golluscio et al. address one of the mechanisms of IKs (KCNQ1/KCNE1) channel upregulation by polyunsaturated fatty acids (PUFAs). PUFAs are known to upregulate KCNQ1 and KCNQ1/KCNE1 channels through two mechanisms: one shifts the voltage dependence in a

negative direction, and the other increases the maximum conductance (G_{max}). While the first mechanism is known to affect the voltage sensor equilibrium through a charge effect, the second mechanism is less understood. Using single-channel recordings and mutagenesis at putative PUFA binding sites, they successfully demonstrate that the selectivity filter is stabilized in a conducting state by PUFA binding, and that this is the mechanism by which PUFAs increase G_{max} . Their single-channel recordings are straightforward and clearly show that the selectivity filter tends to become conductive upon PUFA binding. Since PUFAs are potential therapeutic reagents for cardiac arrhythmias such as long QT syndrome, their findings are beneficial for future research and applications of these compounds.

<https://doi.org/10.7554/eLife.95852.3.sa2>

Reviewer #3 (Public review):

Summary:

This manuscript reveals an important mechanism of KCNQ1/IKs channel gating such that the open state of the pore is unstable and undergoes intermittent closed and open conformations. PUFA enhances the maximum open probability of IKs by binding to a crevice adjacent to the pore and stabilize the open conformation. This mechanism is supported by convincing single channel recordings that show empty and open channel traces and the ratio of such traces is affected by PUFA. In addition, mutations of the pore residues alter PUFA effects, convincingly supporting that PUFA alters the interactions among these pore residues.

Strengths:

The data are of high quality and the description is clear.

<https://doi.org/10.7554/eLife.95852.3.sa1>

Author response:

The following is the authors' response to the previous reviews.

Reviewer #1 (Recommendations For The Authors):

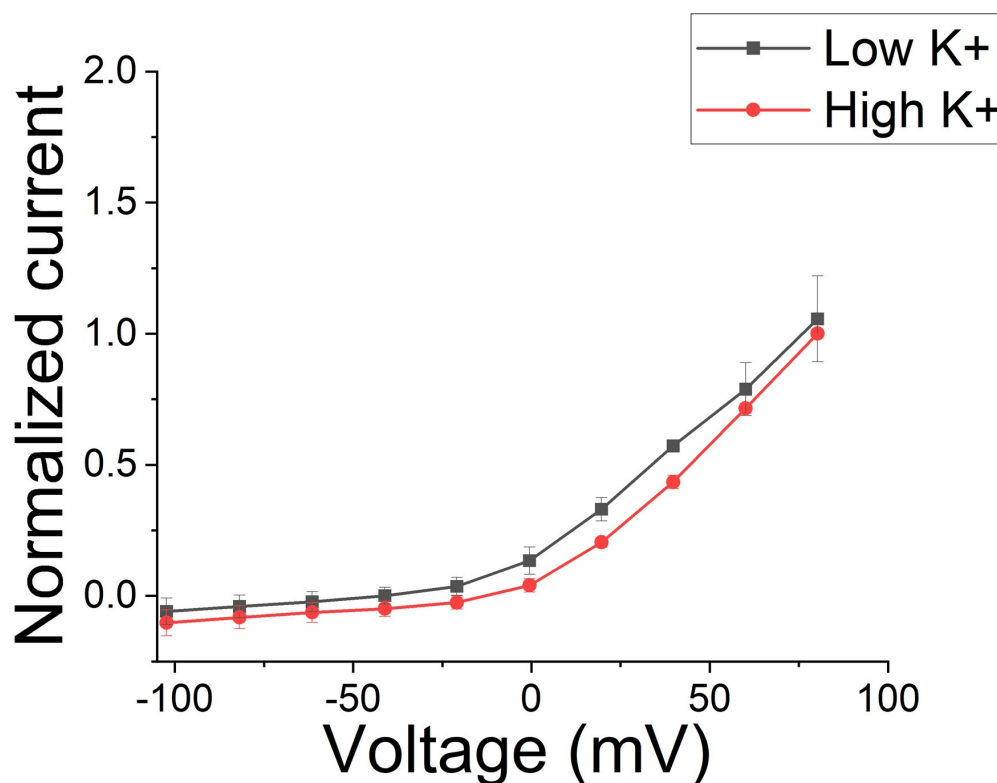
The additional data included in this revision nicely strengthens the major claim.

I apologize that my comment about K^+ concentration in the prior review was unclear. The cryoEM structure of KCNQ1 with S4 in the resting state was obtained with lowered K^+ relative to the active state. Throughout the results and discussion it seems implied that the change in voltage sensor state is somehow causative of the change in selectivity filter state while the paper that identified the structures attributes the change in selectivity filter state not to voltage sensors, but to the change in $[K^+]$ between the 2 structures. Unless there is a flaw in my understanding of the conditions in which the selectivity filter structures used in modeling were generated, it seems misleading to ignore the change in $[K^+]$ when referring to the activated vs resting or up vs down structures. My understanding is that the closed conformation adopted in the resting/low $[K^+]$ is similar to that observed in low $[K^+]$ previously and is more commonly associated with $[K^+]$ -dependent inactivation, not resulting from voltage sensor deactivation as implied here. The original article presenting the low $[K^+]$ structure also suggests this. When discussing conformational changes in the selectivity filter, I strongly suggest referring to these structures as activated/high $[K^+]$ vs resting/low $[K^+]$ or something similar, as the $[K^+]$ concentration is a salient variable.

There seems to be some major confusion here and we will try to explain how we think. Note that in the Mandela and MacKinnon paper, there is no significant difference in the amino acid positions in the selectivity filter between low and high K⁺ when S4 is in the activated position (See Mandala and Mackinnon, PNAS Suppl. Fig S5 C and D). There are only fewer K⁺ in the selectivity filter in low K⁺. So, the structure with the distorted selectivity filter is not due to low K⁺ by itself. Note that there is no real difference between macroscopic currents recorded in low and high K⁺ solutions (except what is expected from changes in driving force) for KCNQ1/KCNE1 channels (Larsen et al., Bioph J 2011), suggesting that low K⁺ do not promote the non-conductive state (Figure 1). We now include a section in the Discussion about high/low K⁺ in the structures and the absence of effects of K⁺ on the function of KCNQ1/KCNE1 channels.

Author response image 1.

Macroscopic KCNQ1/KCNE1 currents recorded in different K⁺ conditions. Note that there is no difference between current recorded in low K⁺ (2 mM) conditions and high (96 mM) K⁺ conditions (n=3 oocytes). Currents were normalized in respect to high K⁺.



Note also that, in the previous version of the manuscript, we did not propose that the position of S4 is what determines the state of the selectivity filter. We only reported that the CryoEM structure with S4 resting shows a distorted selectivity filter. It seems like our text confused the reviewer to think that we proposed that S4 determines the state of the selectivity filter, when we did not propose this earlier. We previously did not want to speculate too much about this, but we have now included a section in the Discussion to make our view clear in light of the confusion of the reviewers.

It is clear from our data that the majority of sweeps are empty (which we assume is with S4 up), suggesting that the selectivity filter can be (and is in the majority of sweeps) in the non-

conducting state even with S4 up. We think that the selectivity filter switches between a non-conductive and a conductive conformation both with S4 down and with S4 up. The cryoEM structure in low K⁺ and S4 down just happened to catch the non-conductive state of the selectivity filter. We have now added a section in the Discussion to clarify all this and explain how we think it works.

However, S4 in the active conformation seems to stabilize the conductive conformation of the selectivity filter, because during long pulses the channel seems to stay open once opened (See Suppl Fig S2). So, one possibility is that the selectivity filter goes more readily into the non-conductive state when S4 is down (and maybe, or not, low K⁺ plays a role) and then when S4 moves up the selectivity filter sometimes recovers into the conductive state and stays there. We now have included a section in the Discussion to present our view. Since this whole discussion was initiated and pushed by the reviewer, we hope that the reviewers will not demand more data to support these ideas. We think that this addition makes sense since other readers might have the same questions and ideas as the reviewer, and we would like to prevent any confusion about this topic.

Figure 1

It remains unclear in the manuscript itself what "control" refers to. Are control patched the same patches that later receive LG?

Yes, the control means the same patch before LG. We now indicate that in legends and text throughout.

Supplementary Figure S1

Unclear if any changes occur after addition of LG in left panel and if the LG data on right is paired in any way to data on left.

Yes, in all cases the left and right panel in all figures are from the same patch. We now indicate that in legends and text throughout.

The letter p is used both to represent open probability open probability from the all-point amplitude histogram and as a p-value statistical probability indicator sometime lower case, sometimes upper case. This was confusing.

We have now exclusively use lower case p for statistical probability and Po for open probability.

"This indicates that mutations of residues in the more intracellular region of the selectivity filter do not affect the Gmax increases and that the interactions that stabilize the channel involve only residues located near the external region part of the selectivity filter. "

Seems too strongly worded, it remains possible that mutations of other residues in the more intracellular region of the selectivity filter could affect the Gmax increases.

We have changed the text to: "Mutations of residues in the more intracellular region of the selectivity filter do not affect the Gmax increases, as if the interactions that stabilize the channel involve residues located near the external region part of the selectivity filter. "

Supplementary Figure S7

Please report Boltzmann fit parameters. What are "normalized" uA?

We removed the uA, which was mistakenly inserted. The lines in the graphs are just lines connecting the dots and not Boltzmann fits, since we don't have saturating curves in all panels to make unique fits.

"We have previously shown that the effects of PUFAs on IKs channels involve the binding of PUFAs to two independent sites." Was binding to the sites actually shown? Suggest changing to: "We have previously proposed models in which the effects of PUFAs..."

We have now changed this as the Reviewer suggested: " We have previously proposed models in which the effects of PUFAs on IKs channels involve the binding of PUFAs to two independent sites."

Statistics used not always clear. Methods refer to multiple statistical tests but it is not clear which is used when.

We use two different tests and it is now explained in figure legends when either was used.

n values confusing. Sometimes # of sweeps used as n. Sometimes # patches used as n. In one instance "The average current during the single channel sweeps was increased by 2.3 {plus minus} 0.33 times (n = 4 patches, p =0.0006)" ...this seems a low p value for this n=4 sample?

We have now more clearly indicated what n stands for in each case. There was an extra 0 in the p value, so now it is $p = 0.006$. Thanks for catching that error.

Reviewer #2 (Recommendations For The Authors):

I still have some comments for the revised manuscript.

(1) (From the previous minor point #6) Since D317E and T309S did not show statistical significance in Figure 5A, the sentences such as "This data shows that Y315 and D317 are necessary for the ability of Lin-Glycine to increase Gmax" or "the effect of Lin-Glycine on Gmax of the KCNQ1/KCNE1 mutant was noticeably reduced compared to the WT channel showing the this residue contributes to the Gmax effect (Figure 5A)." may need to be toned down. Alternatively, I suggest the authors refer to Supplementary Figure S7 to confirm that Y315 and D317 are critical for increasing Gmax.

We have redone the analysis and statistical evaluation in Fig 5. We now use the more appropriate value of the fitted Gmax (which use the whole dose response curve instead of only the 20 mM value) in the statistical evaluation and now Y315F and D317E are statistically different from wt.

(2) Supplementary Fig. S1. All control diary plots include the green arrows to indicate the timing of lin-glycine (LG) application. It is a bit confusing why they are included. Is it to show that LG application did not have an immediate effect? Are the LG-free plots not available?

Not sure what the Reviewer is asking about? In the previous review round the Reviewers asked specifically for this. The arrow shows when LG was applied and the plot on the right shows the effect of LG from the same patch.

(3) The legend to Supplementary Figure S4, "The side chain of residues ... are highlighted as sticks and colored based on the atomic displacement values, from white to blue to red on a scale of 0 to 9 Å." They look mostly blue (or light blue). Which one is colored white? It

might be better to use a different color code. It would also be nice to link the color code to the colors of Supplementary Figure S5, which currently uses a single color.

We have removed “from white to blue to red on a scale of 0 to 9 Å” and instead now include a color scale directly in Fig S4 to show how much each atom moved based on the color.

We feel it is not necessary to include color in Fig S5 since the scale of how much each atom moves is shown on the y axis.

(4) Add unit (pA) to the y-axis of Supplementary Figure S2.

pA has been added.

Reviewer #3 (Recommendations For The Authors):

Some issues on how data support conclusions are identified. Further justifications are suggested.

186: “The decrease in first latency is most likely due to an effect of Lin-Glycine on Site I in the VSD and related to the shift in voltage dependence caused by Lin-Glycine.” The results in Fig S1B do not seem to support this statement since the mutation Y315F in the pore helix seemed to have eliminated the effect of Lin-Glycine in reducing first latency. The authors may want to show that a mutation that eliminating Site I would eliminate the effect of Lin-Glycine on first latency. On the other hand, it will be also interesting to examine if another pore mutation, such as P320L (Fig 5) also reduce the effect of Lin-Glycine on first latency.

These experiments are very hard and laborious, and we feel these are outside the scope of this paper which focuses on Site II and the mechanism of increasing Gmax. Further studies of the voltage shift and latency will have to be for a future study.

The mutation D317E did not affect the effect of Lin-Glycine on Gmax significantly (Fig 5A, and Fig S7F comparing with Fig S7A), but the authors conclude that D317 is important for Lin-Glycine association. This conclusion needs a better justification.

We have redone the analysis and statistical evaluation in Fig 5. We now use the more appropriate value of the fitted Gmax (which use the whole dose response curve instead of only the 20 mM value) in the statistical evaluation and now D317E is statistically different from wt

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